

Flatness-Based and Passivity-Based Control with Adaptive Neural Compensation of a Direct-Drive PMSG Wind Energy Conversion System

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Abstract—This paper proposes a control strategy for a grid-connected 2 MW direct-drive permanent magnet synchronous generator (PMSG) wind turbine with a back-to-back converter and an L filter. The outer loops are designed by differential flatness: the rotor speed is the flat output of the maximum power point tracking (MPPT) loop, and the energy stored in the DC link and in the filter is the flat output of the DC-link loop. The inner current loops of both converters use passivity-based control (PBC), which keeps the workless coupling of the machine and filter models and injects damping; a storage function proves their stability. An adaptive 6-10-2 neural network, updated online with dead-zone, σ -modification and back-propagation, adds a compensation signal to the flat loops without changing the nominal gains. Simulations under turbulent wind, parameter mismatch, a 10 % error on the aerodynamic model and a 20 % grid voltage dip are compared with PI vector control. The proposed control tracks the MPPT speed 3.8 times more accurately and limits the DC-link deviation during the dip to 3.6 V, against 11.6 V with PI and power feedforward, while the neural compensation reduces the integral absolute speed error by 38 %.

Index Terms—PMSG, Wind Energy Conversion Systems, Back-to-Back Converter, Flatness-Based Control, Passivity-Based Control, Adaptive Neural Network, Maximum Power Point Tracking (MPPT).

I. INTRODUCTION

Wind energy is one of the main sources in the transition towards low-carbon electricity generation [1]; a record 117 GW of new wind capacity was installed worldwide in 2023 [2]. Direct-drive PMSG turbines remove the gearbox, which reduces maintenance and losses, and their full-scale back-to-back converter decouples the generator from the grid [3], [4]; it must extract the maximum power, regulate the DC-link voltage and inject a clean current.

These tasks are usually performed by cascaded PI loops in rotating frames [5], [6]. The PMSG model is nonlinear and coupled, and its parameters vary with temperature and saturation. Nonlinear controllers have therefore been studied for wind turbines and PMSGs, such as backstepping [7], predictive control [8] and passivity-based control [9].

Differential flatness [10] computes the control input directly from a desired trajectory of a flat output; using the stored energy as the flat output gives a simple linear power balance

for DC buses [11]. Passivity-based control [12], [13] shapes the energy of the closed loop and keeps the physical structure of the system. Both methods rely on a model, so model errors leave residual tracking errors. Adaptive neural networks can learn such residual terms online [14]; in [15], an adaptive neural compensation was added to a nominal controller of a DFIG without changing its gains.

This paper combines these tools for the complete chain of a 2 MW direct-drive PMSG turbine. The individual techniques are known; the contributions are their adaptation to this chain and the resulting analysis:

- a DC-link flat output that includes the energy of the L filter, so that the grid-side law contains the measured generator power as an exact feedforward and the DC-link error dynamics is linear;
- PBC laws for the currents of both converters and a cascade proof that the complete nominal interconnection (flat outer loops and PBC inner loops) is exponentially stable;
- the transfer of this compensation, developed in [15] for the rotor currents of a DFIG, to the two flat loops of a PMSG, with a learning signal taken along their control directions and normalized update laws, for which a direct boundedness proof is given;
- an evaluation against PI vector control with and without power feedforward, over several wind realizations, initial conditions and adaptation settings, and with a switched converter model.

II. WIND TURBINE MODELING

The aerodynamic power captured by the turbine is [17]

$$P_w = \frac{1}{2} \rho \pi R^2 C_p(\lambda, \beta) v^3 \quad (1)$$

where ρ is the air density, R the rotor radius, v the wind speed, β the pitch angle and C_p the power coefficient, which depends on the tip-speed ratio

$$\lambda = \frac{\Omega R}{v} \quad (2)$$

with Ω the rotor speed. The power coefficient is modeled by

$$C_p = 0.5176 \left(\frac{116}{\lambda_i} - 0.4\beta - 5 \right) e^{-21/\lambda_i} + 0.0068\lambda \quad (3)$$

with $1/\lambda_i = 1/(\lambda + 0.08\beta) - 0.035/(\beta^3 + 1)$. For $\beta = 0$, its maximum $C_{p,\max} = 0.48$ is reached at $\lambda_{opt} = 8.1$. The aerodynamic torque is $T_w = P_w/\Omega$, and the direct-drive shaft is described by the one-mass model

$$J \frac{d\Omega}{dt} = T_w - T_e - B\Omega \quad (4)$$

where J is the total inertia, T_e the electromagnetic torque and B the viscous friction coefficient. With $R = 40$ m and $\Omega_n = 2.23$ rad/s, the rated wind speed is $v_n = \Omega_n R / \lambda_{opt} \approx 11$ m/s and the captured power at v_n is about 2 MW.

III. PMSG AND BACK-TO-BACK CONVERTER MODEL

A. PMSG

In the rotor dq frame, with the d axis aligned with the magnet flux, generator convention and $L_d = L_q = L_s$, the PMSG is described by [4]

$$L_s \frac{di_{sd}}{dt} = -R_s i_{sd} + \omega_e L_s i_{sq} - v_{sd} \quad (5)$$

$$L_s \frac{di_{sq}}{dt} = -R_s i_{sq} - \omega_e L_s i_{sd} + \omega_e \psi_f - v_{sq} \quad (6)$$

where $\omega_e = p\Omega$, p is the number of pole pairs and ψ_f the magnet flux. The electromagnetic torque is

$$T_e = \frac{3}{2} p \psi_f i_{sq}. \quad (7)$$

With $\mathbf{i}_s = [i_{sd} \ i_{sq}]^T$, $\mathbf{v}_s = [v_{sd} \ v_{sq}]^T$, $\mathbf{e}_q = [0 \ 1]^T$ and the skew-symmetric matrix $\mathbf{J} = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$, the model reads

$$L_s \dot{\mathbf{i}}_s = -R_s \mathbf{i}_s + \omega_e L_s \mathbf{J} \mathbf{i}_s + \omega_e \psi_f \mathbf{e}_q - \mathbf{v}_s. \quad (8)$$

Since $\mathbf{i}_s^T \mathbf{J} \mathbf{i}_s = 0$, the coupling term $\omega_e L_s \mathbf{J} \mathbf{i}_s$ is workless: it exchanges energy between the two axes without producing or dissipating it.

B. Back-to-Back Converter, DC Link and Grid

The two-level converters are represented by their average model. The voltages they synthesize are limited by the space-vector modulation range, $\|\mathbf{v}_s\|, \|\mathbf{v}_i\| \leq V_{dc}/\sqrt{3}$, and, neglecting losses, the DC-link capacitor obeys

$$CV_{dc} \frac{dV_{dc}}{dt} = P_{msc} - P_{gsc} \quad (9)$$

with $P_{msc} = \frac{3}{2} \mathbf{v}_s^T \mathbf{i}_s$ and $P_{gsc} = \frac{3}{2} \mathbf{v}_i^T \mathbf{i}_g$. In the grid-voltage-oriented frame ($v_{gd} = V_m$, $v_{gq} = 0$), the L filter gives

$$L_f \dot{\mathbf{i}}_g = -R_f \mathbf{i}_g + \omega_g L_f \mathbf{J} \mathbf{i}_g + \mathbf{v}_i - \mathbf{v}_g \quad (10)$$

and the powers injected into the grid are $P = \frac{3}{2} v_{gd} i_{gd}$ and $Q = -\frac{3}{2} v_{gd} i_{gq}$.

TABLE I
PARAMETERS OF THE STUDIED SYSTEM

Parameter	Value	Parameter	Value
P_n	2 MW	p	26
R	40 m	R_s	0.78 mΩ
Ω_n	2.23 rad/s	L_s	1.57 mH
T_n	0.897 MN·m	ψ_f	9.18 Wb
$\lambda_{opt}, C_{p,\max}$	8.1, 0.48	C	20 mF
ρ	1.225 kg/m ³	V_{dc}^*	1200 V
J	2.8×10^6 kg·m ²	L_f, R_f	0.15 mH, 2 mΩ
B	10^3 N·m·s/rad	Grid	690 V, 50 Hz

C. Complete Model

With the state $\mathbf{x} = [i_{sd} \ i_{sq} \ \Omega \ V_{dc} \ i_{gd} \ i_{gq}]^T$, the inputs $\mathbf{u} = [\mathbf{v}_s^T \ \mathbf{v}_i^T]^T$ and the disturbances v and $\mathbf{v}_g$, (4), (8), (9) and (10) form a sixth-order nonlinear model, denoted $\dot{\mathbf{x}} = f(\mathbf{x}, \mathbf{u}, v, \mathbf{v}_g)$ in the following. The parameters are listed in Table I. The turbine data correspond to a standard 2 MW turbine; the generator, DC-link, filter and grid data are those of the non-salient PMSG of [16]. The magnet flux is consistent with the grid: at 9.75 Hz, the rated frequency of [16], the back-EMF $\psi_f \omega_e = 562$ V equals the peak phase voltage of the 690 V grid.

IV. WIND PROFILE AND REFERENCE GENERATION

The wind speed combines a mean value, low-frequency harmonics and a turbulence term [15]:

$$v(t) = V_{mean} + \sum_{k=1}^3 A_k \sin(2\pi f_k t) + v_{turb}(t) \quad (11)$$

with $V_{mean} = 8$ m/s, $A = [1.5, 0.2, 0.3]$ m/s, $f = [0.03, 0.16, 0.17]$ Hz, and v_{turb} a smoothed Gaussian noise (standard deviation 0.01 m/s, time constant 0.05 s).

The maximum power is extracted at $\lambda = \lambda_{opt}$. The measured wind is filtered, $\tau_v \dot{\hat{v}} = v - \hat{v}$ with $\tau_v = 0.3$ s, and the speed reference is obtained from the critically damped filter

$$\ddot{\Omega}^* + 2\omega_r \dot{\Omega}^* + \omega_r^2 \Omega^* = \omega_r^2 \frac{\lambda_{opt} \hat{v}}{R} \quad (12)$$

with $\omega_r = 1.5$ rad/s, which provides Ω^* and $\dot{\Omega}^*$ without numerical differentiation. Since T_e only depends on i_{sq} , $i_{sd}^* = 0$; unity power factor gives $i_{gq}^* = 0$; the DC-link reference is $V_{dc}^* = 1200$ V.

V. PROPOSED CONTROL

The control structure is shown in Fig. 1. It is sampled at $T_s = 100 \mu\text{s}$.

A. Flatness-Based Outer Loops

A system is flat if its states and inputs can be written as functions of a flat output and of a finite number of its derivatives [10]. The control is then obtained by inverting the model with a desired dynamics imposed on the flat output. Flatness is used here only for the two outer subsystems, in which the currents are treated as inputs: the shaft equation (4) with input i_{sq} , and the power balance of the DC link and

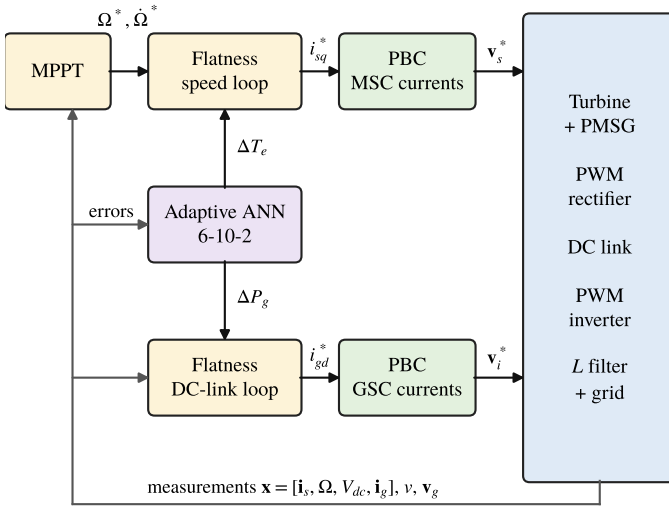


Fig. 1. Proposed control: flat outer loops, PBC current loops and adaptive neural compensation.

filter with input i_{gd} . In both, T_w (or its estimate), P_{msc} and i_{gq} are known exogenous signals. No flatness property is claimed for the complete sixth-order model with the converter voltages as inputs; the coupling between the outer and inner loops is treated by Proposition 2.

Speed loop. From (4), $T_e = T_w - B\Omega - J\dot{\Omega}$, so $y_1 = \Omega$ is a flat output. Replacing $\dot{\Omega}$ by

$$\nu_1 = \dot{\Omega}^* + k_{11}e_\Omega + k_{12} \int e_\Omega dt, \quad e_\Omega = \Omega^* - \Omega \quad (13)$$

gives

$$T_e^* = \hat{T}_w - B\Omega - J\nu_1 + \Delta T_e, \quad i_{sq}^* = \frac{T_e^*}{\frac{3}{2}p\psi_f} \quad (14)$$

where $\hat{T}_w$ is computed from (1)–(3) with the measured wind and ΔT_e is the neural compensation. T_e^* is limited to $[0, 1.2T_n]$ and the integral is frozen during saturation. If $\hat{T}_w = T_w$ and $\Delta T_e = 0$, the error obeys $\dot{e}_\Omega + k_{11}e_\Omega + k_{12} \int e_\Omega dt = 0$.

DC-link loop. The flat output is the energy stored in the DC link and in the filter,

$$y_2 = \frac{1}{2}CV_{dc}^2 + \frac{3}{4}L_f\|i_g\|^2. \quad (15)$$

From (9) and (10), using $i_g^T J i_g = 0$,

$$\dot{y}_2 = P_{msc} - \frac{3}{2}v_{gd}i_{gd} - \frac{3}{2}R_f\|i_g\|^2 \quad (16)$$

which is linear in the injected power $P_g = \frac{3}{2}v_{gd}i_{gd}$, as in [11]. Taking y_2 as the state of this first-order energy balance with input P_g , the input is $P_g = P_{msc} - \frac{3}{2}R_f\|i_g\|^2 - \dot{y}_2$ and V_{dc} follows from y_2 and the measured currents, so y_2 is a flat output of the reduced subsystem. With $e_y = y_2^* - y_2$ and $\dot{y}_2^* \approx 0$, imposing $\dot{e}_y = -\nu_2$, i.e. $\dot{y}_2 = \nu_2$, with $\nu_2 = k_{21}e_y + k_{22} \int e_y dt$, yields

$$P_g^* = P_{msc} - \frac{3}{2}R_f\|i_g\|^2 - \nu_2 + \Delta P_g, \quad i_{gd}^* = \frac{2P_g^*}{3v_{gd}}. \quad (17)$$

Inserting (17) into the expression of $\dot{y}_2$ with $i_{gd} = i_{gd}^*$ gives $\dot{y}_2 = \nu_2 - \Delta P_g$, hence $\dot{e}_y + k_{21}e_y + k_{22} \int e_y dt = \Delta P_g$. The term P_{msc} , known from the applied voltages and measured currents, anticipates the power coming from the generator. The reference $y_2^* = \frac{1}{2}CV_{dc}^{*2} + \frac{3}{4}L_f\|i_g^*\|^2$ varies only through the filter energy (at most 300 J, against 14.4 kJ in the capacitor), and $\dot{y}_2^* = \frac{3}{2}L_f i_g^{*T} \dot{i}_g^*$ is neglected. It enters (26) below as an input, and the impulse response of $s/(s^2 + k_{21}s + k_{22})$ has an L_1 norm of $2/(e\omega_v) = 12.3$ ms, so its effect is $|e_y| \leq 12.3 \text{ ms} \cdot \sup |\dot{y}_2^*|$. In simulation, $|\dot{y}_2^*| \leq 2.7 \text{ kW}$ (0.14 % of P_n) outside the transients, i.e. less than 34 J or 1.4 V on V_{dc} ; it reaches 29 kW at start-up and 52 kW during the grid dip. Adding $\dot{y}_2^*$ to ν_2 reduces the ISE of V_{dc} by 3 % but raises the dip deviation from 3.6 to 5.3 V (numerical differentiation of i_g^*), so it is omitted. The gains are $k_{11} = 2\zeta_\Omega\omega_\Omega$, $k_{12} = \omega_\Omega^2$, $k_{21} = 2\zeta_v\omega_v$, $k_{22} = \omega_v^2$ (Table II).

B. Passivity-Based Current Loops

For the machine-side converter, the error $e_s = i_s - i_s^*$ and its integral z_s ($\dot{z}_s = e_s$) are used in

$$v_s^* = -L_s \dot{i}_s^* - R_s i_s^* + \omega_e L_s J i_s^* + \omega_e \psi_f e_q + R_a e_s + K_i z_s \quad (18)$$

where R_a is the injected damping. For the grid-side converter, with $e_g = i_g - i_g^*$ and $\dot{z}_g = e_g$,

$$v_i^* = L_f \dot{i}_g^* + R_f i_g^* - \omega_g L_f J i_g^* + v_g - R_b e_g - K_{ig} z_g. \quad (19)$$

The reference derivatives are obtained by a first-order filter ($\tau_d = 2$ ms) and the voltages are limited to $V_{dc}/\sqrt{3}$, with the integrals frozen at the limit.

Proposition 1. With (18) applied to (8) and $R_a, K_i > 0$, the storage function $H = \frac{1}{2}L_s\|e_s\|^2 + \frac{1}{2}K_i\|z_s\|^2$ satisfies $\dot{H} = -(R_s + R_a)\|e_s\|^2 \leq 0$, and $e_s \rightarrow 0$. The same holds for (19) applied to (10).

Proof. Subtracting $L_s \dot{i}_s^*$ from (8) and inserting (18) gives $L_s \dot{e}_s = -(R_s + R_a)e_s + \omega_e L_s J e_s - K_i z_s$. Hence $\dot{H} = e_s^T L_s \dot{e}_s + K_i z_s^T e_s = -(R_s + R_a)\|e_s\|^2 + \omega_e L_s e_s^T J e_s$, and the last term vanishes because J is skew-symmetric. H is bounded and $\dot{H}$ is uniformly continuous, so Barbalat's lemma [18] gives $e_s \rightarrow 0$. $\square$

C. Adaptive Neural Compensation

As in [15], the network does not replace the nominal control u_N given by (14)–(17) and does not modify its gains; it adds

$$u = u_N + u_{AI}, \quad u_{AI} = [\Delta T_e \ \Delta P_g]^T. \quad (20)$$

Its six inputs are built from the errors $\bar{e} = [e_\Omega/(0.01\Omega_n), e_y/(300 \text{ J}), e_{isq}/(0.01I_n)]^T$, normalized by 1 % of the rated speed, 300 J (about 1 % of V_{dc}^*) and 1 % of the rated current $I_n = T_n/(\frac{3}{2}p\psi_f)$, and from their derivatives, filtered with a 5 ms time constant and scaled by $\tau = [0.5, 0.02, 0.002] \text{ s}$: $z_n = [\bar{e}^T, (\tau \odot \bar{e})^T]^T$. With one hidden layer of ten neurons,

$$h = \tanh(W_1 z_n + b_1), \quad y = W_2 h + b_2, \quad (21)$$

$$u_{AI} = K_{AI} \text{diag}(0.1T_n, 0.1P_n) \text{sat}_{\pm 3}(y) \quad (22)$$

with $W_1 \in \mathbb{R}^{10 \times 6}$ and $W_2 \in \mathbb{R}^{2 \times 10}$. T_e and P_g enter $\dot{e}_\Omega$ and $\dot{e}_y$ with a positive sign, so the learning signal in the control direction is $\mathbf{r} = [\bar{e}_1 \ \bar{e}_2]^T$, set to zero inside a dead zone $\|\mathbf{r}\| \leq \delta$. The parameters are updated online by

$$\dot{W}_2 = -\Gamma_n \mathbf{r} \mathbf{h}^T - \sigma W_2, \quad \dot{\mathbf{b}}_2 = -\Gamma_n \mathbf{r} - \sigma \mathbf{b}_2, \quad (23)$$

$$\dot{W}_1 = -\Gamma_n \delta_h \mathbf{z}_n^T - \sigma W_1, \quad \dot{\mathbf{b}}_1 = -\Gamma_n \delta_h - \sigma \mathbf{b}_1, \quad (24)$$

with $\delta_h = (1 - \mathbf{h}^2) \odot (W_2^T \mathbf{r})$ and the normalized gain $\Gamma_n = \Gamma / (1 + \|\mathbf{z}_n\|^2)$. The σ -modification terms [14], [19] prevent parameter drift, and the normalization [19] slows the adaptation when the errors are large, e.g. after a torque saturation; with a constant gain, a start 10 % below the MPPT speed led to an oscillatory adaptation in simulation. Compared with [15], this normalization is the only change to the update laws. The adaptation is frozen when T_e^* or i_{gd}^* saturates, and u_{AI} is applied at the next sample. W_2 and $\mathbf{b}_2$ start at zero, so the compensation is initially inactive.

Proposition 2. *Consider the model of Section III-C with (14)–(19), $\Delta T_e = \Delta P_g = 0$, an exact model, no saturation, exact reference derivatives, $\dot{y}_2^* = 0$ and $|\omega_e| \leq \bar{\omega}$ on the operating region. Then the origin of the error state $\boldsymbol{\xi} = (\mathbf{e}_s, \mathbf{z}_s, \mathbf{e}_g, \mathbf{z}_g, e_\Omega, \int e_\Omega, e_y, \int e_y)$ is exponentially stable.*

Proof. Inner loops. With $V_s = H + \varepsilon \mathbf{e}_s^T L_s \mathbf{z}_s$, $\dot{V}_s = -(R_s + R_a)\|\mathbf{e}_s\|^2 + \varepsilon(L_s\|\mathbf{e}_s\|^2 - (R_s + R_a)\mathbf{e}_s^T \mathbf{z}_s - \omega_e L_s \mathbf{e}_s^T \mathbf{J} \mathbf{z}_s - K_i \|\mathbf{z}_s\|^2)$. Since $|\omega_e| \leq \bar{\omega}$, a small $\varepsilon > 0$ makes V_s positive definite and $\dot{V}_s \leq -c(\|\mathbf{e}_s\|^2 + \|\mathbf{z}_s\|^2)$ with $c > 0$, so $(\mathbf{e}_s, \mathbf{z}_s) \rightarrow 0$ exponentially; the same holds for $(\mathbf{e}_g, \mathbf{z}_g)$ since ω_g is constant. *Outer loops.* With the actual currents, $T_e = T_e^* + \frac{3}{2}p\psi_f e_{sq}$ and $i_{gd} = i_{gd}^* + e_{gd}$, so (4) and the y_2 dynamics give

$$\begin{aligned} \dot{e}_\Omega + k_{11}e_\Omega + k_{12}\int e_\Omega &= \frac{3p\psi_f}{2J} e_{sq}, \\ \dot{e}_y + k_{21}e_y + k_{22}\int e_y &= \frac{3}{2}v_{gd} e_{gd}. \end{aligned}$$

These are exponentially stable linear systems ($k_{ij} > 0$) driven by exponentially decaying inputs with bounded gains. The inner error dynamics do not depend on the outer states, so the interconnection is a cascade of exponentially stable systems and is exponentially stable [18]. $\square$

The boundedness proof of [15] uses a Lyapunov function with $\text{tr}(\tilde{W}^T \Gamma^{-1} \tilde{W})$ and a *constant* gain; with $\Gamma_n(t)$ an additional term in $\tilde{\Gamma}_n$ appears, so it does not apply directly, but the normalization gives a direct bound. The model error $\Delta = T_w - \hat{T}_w$ and the compensation enter the outer error equations of Proposition 2 as additional inputs:

$$\dot{e}_\Omega + k_{11}e_\Omega + k_{12}\int e_\Omega = \frac{3p\psi_f}{2J} e_{sq} + \frac{1}{J}(\Delta T_e - \Delta), \quad (25)$$

$$\dot{e}_y + k_{21}e_y + k_{22}\int e_y = \frac{3}{2}v_{gd} e_{gd} + \Delta P_g + \dot{y}_2^*. \quad (26)$$

Proposition 3. *Let $\Gamma, \sigma > 0$. (i) Whatever the tracking errors, the parameters of the network remain bounded: with Frobenius norms, $\|W_2(t)\| \leq$*

TABLE II
CONTROLLER AND ADAPTIVE ANN PARAMETERS

Parameter	Value
Speed loop	$\omega_\Omega = 2 \text{ rad/s}$, $\zeta_\Omega = 0.9$ ($k_{11} = 3.6$, $k_{12} = 4$)
DC-link loop	$\omega_v = 60 \text{ rad/s}$, $\zeta_v = 1$ ($k_{21} = 120$, $k_{22} = 3600$)
PBC (MSC)	$R_a = 0.785 \ \Omega$, $K_i = 98.1 \ \Omega/\text{s}$
PBC (GSC)	$R_b = 0.075 \ \Omega$, $K_{ig} = 9.38 \ \Omega/\text{s}$
ANN architecture	6–10–2
$\Gamma, \sigma, \delta, K_{AI}$	5, 0.01, 0.1, 1
Output saturation	± 3 ($\pm 0.3T_n$, $\pm 0.3P_n$)

$\bar{W}_2 := \max(\|W_2(0)\|, \sqrt{10}\Gamma/(2\sigma))$, $\|\mathbf{b}_2(t)\| \leq \max(\|\mathbf{b}_2(0)\|, \Gamma/(2\sigma))$, $\|W_1(t)\| \leq \max(\|W_1(0)\|, \Gamma\bar{W}_2/\sigma)$ and $\|\mathbf{b}_1(t)\| \leq \max(\|\mathbf{b}_1(0)\|, \Gamma\bar{W}_2/(2\sigma))$. (ii) Under the assumptions of Proposition 2, except $\Delta T_e = \Delta P_g = 0$, and for bounded Δ and $\dot{y}_2^*$, the error state $\boldsymbol{\xi}$ is uniformly ultimately bounded, with an ultimate bound proportional to $\sup_t(|\Delta T_e - \Delta|/J + |\Delta P_g + \dot{y}_2^*|)$, as long as the trajectory remains in the operating region of Proposition 2.

Proof. (i) The learning signal $\mathbf{r}$ is zero or equal to the first two entries of $\mathbf{z}_n$, so $\|\mathbf{r}\| \leq \|\mathbf{z}_n\|$, and $\|\mathbf{h}\| \leq \sqrt{10}$. Hence $\Gamma_n \|\mathbf{r}\| \|\mathbf{h}\| \leq \sqrt{10}\Gamma \|\mathbf{z}_n\|/(1 + \|\mathbf{z}_n\|^2) \leq \sqrt{10}\Gamma/2$, and $\frac{d}{dt} \frac{1}{2} \|W_2\|^2 \leq -\sigma \|W_2\|^2 + \frac{\sqrt{10}}{2} \Gamma \|W_2\|$, which gives the bound on W_2 ; that on $\mathbf{b}_2$ follows in the same way. Then $\|\delta_h\| \leq \bar{W}_2 \|\mathbf{r}\|$ and $\Gamma_n \|\delta_h\| \|\mathbf{z}_n\| \leq \Gamma \bar{W}_2 \|\mathbf{z}_n\|^2/(1 + \|\mathbf{z}_n\|^2) \leq \Gamma \bar{W}_2$, which bounds W_1 and $\mathbf{b}_1$. Freezing the adaptation keeps the parameters constant. (ii) u_{AI} is bounded by its saturation ($\pm 0.3T_n$, $\pm 0.3P_n$). The error system of Proposition 2 is exponentially stable and the inputs of (25)–(26) enter it linearly, so it is locally input-to-state stable [18], which gives the bound. $\square$

Proposition 3 holds for any $\Gamma > 0$, independently of the approximation capability of the network, and does not rely on [15]. It does not prove that ΔT_e approaches Δ , i.e., that the network reduces the bound; this is assessed in Section VI.

The bounds grow with Γ/σ : in simulation, $\Gamma \geq 20$ saturates the network output and the DC-link loop oscillates with a bounded amplitude; $\Gamma = 5$ is used.

VI. NUMERICAL SIMULATIONS

A. Setup

The model of Section III-C is integrated with a fourth-order Runge–Kutta scheme at step T_s over 10 s, starting from the MPPT operating point at $v = 8 \text{ m/s}$. Two cases are considered:

- *nominal*: the controller uses the true parameters;
- *robustness*: the controller keeps the nominal values while the plant has $R_s \times 1.5$, $L_s \times 1.2$, $\psi_f \times 0.92$, $R_f \times 1.5$, $L_f \times 1.25$; $\hat{T}_w$ is 10 % lower than T_w ; and the grid voltage drops by 20 % between 8 and 8.2 s.

The proposed control, without and with the network, is compared with PI vector control (speed, DC-voltage and dq -decoupled current loops [5], [6]) tuned on the same poles, with and without the feedforward $P_{msc}/(\frac{3}{2}v_{gd})$ (PI+FF), using the same reference (12) and limits.

TABLE III
COMPARISON OF THE CONTROLLERS (10 s)

Metric	PI	PI+FF	Proposed no ANN	Proposed + ANN
<i>Nominal</i>				
IAE of Ω (rad)	0.0384	0.0384	2.7×10^{-6}	2.7×10^{-6}
ISE of V_{dc} (V ² s)	26.4	0.095	0.085	0.085
Max $ V_{dc} - V_{dc}^* $ (V)	2.87	2.46	0.36	0.36
Energy (kWh)	2.612	2.612	2.625	2.625
<i>Robustness</i>				
IAE of Ω (rad)	0.0419	0.0419	0.0109	0.0068
RMS error of i_{sq} (A)	0.84	0.84	3.05	3.07
RMS error of i_{gd} (A)	2.15	5.98	6.88	7.29
ISE of V_{dc} (V ² s)	255	2.69	1.30	1.25
Max $ V_{dc} - V_{dc}^* $, $t < 1$ s (V)	4.53	3.85	10.44	10.44
Max $ V_{dc} - V_{dc}^* $, dip (V)	62.3	11.6	3.35	3.60
Energy (kWh)	2.599	2.599	2.615	2.614

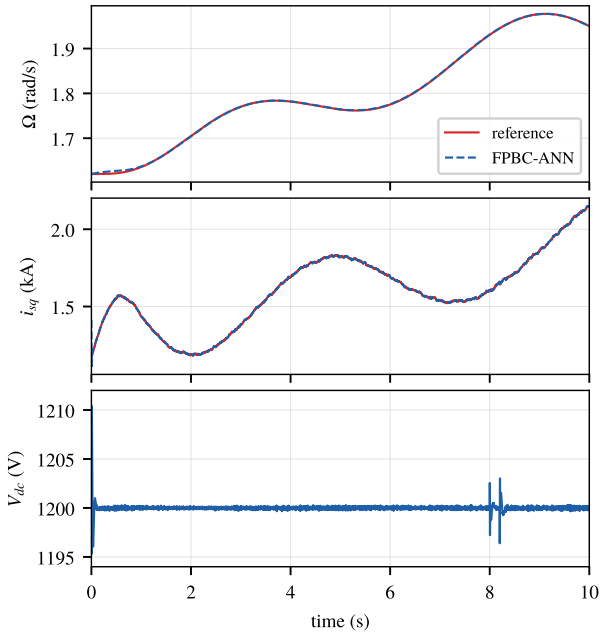


Fig. 2. Robustness case with the proposed control: rotor speed, q -axis generator current and DC-link voltage.

B. Comparison with PI Vector Control

Table III and Figs. 2–3 summarize the results. *Speed*: since the flat law contains $J\dot{\Omega}^*$ and $\hat{T}_w$, the IAE is 3.8 times (robustness) to 10^4 times (nominal) smaller than with PI, and the energy is 0.5–0.6 % higher. *DC link*: the feedforward of $P_{m,sc}$ is decisive (PI without it gives a 200–300 times larger ISE); during the dip, the deviation is 3.6 V, against 11.6 V with PI+FF and 62 V with PI. PI+FF keeps a smaller steady ripple (0.03 vs 0.36 V) and start-up deviation (3.9 vs 10.4 V). *Currents*: PI tracks them better (RMS error of i_{sq} 0.84 vs 3.07 A under mismatch), because the PBC feedforward uses nominal parameters; all errors remain below 0.3 % of the rated current.

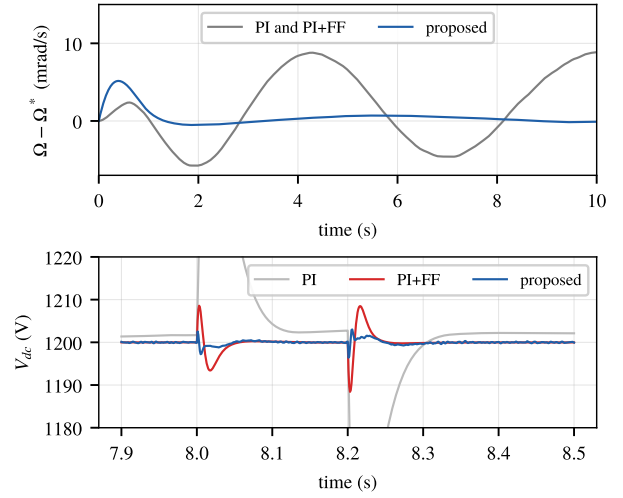


Fig. 3. Robustness case: speed error of the PI controllers and of the proposed control (top), and DC-link voltage during the 20 % grid voltage dip (bottom).

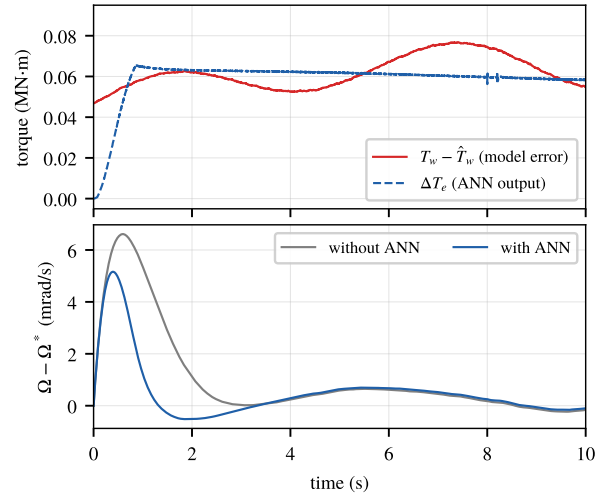


Fig. 4. Robustness case: neural compensation ΔT_e compared with the aerodynamic model error, and speed error with and without the network.

C. Effect of the Neural Compensation

Fig. 4 shows that, without knowing the 10 % model error, the network converges in less than one second to $\Delta T_e \approx 0.065$ MN·m, close to the mean of the actual error $T_w - \hat{T}_w$ (0.047 to 0.077 MN·m); the slowly varying part is handled by the integral action of (14). The IAE of the speed decreases by 38 %, mainly during the start-up transient, where the peak speed error drops from 6.6 to 5.2 mrad/s. Table IV reports additional tests.

Over five turbulence realizations, the IAE is reduced by 38 ± 0.1 % and the ISE of V_{dc} by 4 %; the current errors and the dip deviation increase slightly (Table III). A larger Γ improves the speed tracking at the expense of the current and DC-link errors; a larger dead zone or σ reduces the compensation. Starting 10 % below the MPPT speed, the network reduces the IAE by 10 % with a well-damped response; starting 10 %

TABLE IV
ADDITIONAL TESTS OF THE NEURAL COMPENSATION (ROBUSTNESS
CASE; FIVE WIND REALIZATIONS: MEAN, SPREAD $< 10^{-5}$ RAD AND
0.01 V²S; Ω_0 : INITIAL SPEED)

Test	IAE of Ω (rad)	ISE of V_{dc} (V ² s)	RMS e_{igd} (A)
5 winds, no ANN	0.01093	1.29 ± 0.01	–
5 winds, ANN	0.00678	1.24 ± 0.01	–
$\Gamma = 1$	0.0092	1.28	6.96
$\Gamma = 2$	0.0082	1.27	7.04
$\Gamma = 5$ (used)	0.0068	1.25	7.29
$\Gamma = 10$	0.0061	1.24	7.90
$\Gamma = 20$	oscillatory (V_{dc} deviation 47 V)		
$\delta = 0.05 / 0.2$	0.0071 / 0.0073	1.23 / 1.25	7.24 / 7.13
$\sigma = 0.001 / 0.1$	0.0067 / 0.0083	1.25 / 1.24	7.36 / 7.01
$\Omega_0 = 0.9\Omega^*$: no ANN/ANN	0.0921 / 0.0828	1.15 / 1.34	7.95 / 8.02
$\Omega_0 = 0.9\Omega^*$: PI+FF	0.1142	23.8	14.2

above, all controllers saturate the grid current and the DC link rises by over 120 V, which requires a braking chopper or pitch control.

D. Switched Model and Computational Cost

A switched model of the two-level converters (space-vector modulation, 2160 Hz carrier, double update, 1 μ s step) was simulated at 8, 9 and 10 m/s with nominal parameters. The phase-*a* grid current is analyzed by FFT over ten periods in steady state; as in IEEE 519 [20], the harmonics $h = 2, \dots, 50$ are referred to the rated current I_L at the converter terminals. The total demand distortion (TDD) is 0.12, 0.07 and 0.13 %, and the largest harmonic 0.11 % of I_L , below the limits (5 % TDD, 0.3 % for $35 \leq h \leq 50$ in the strictest class). The switching components are not covered by these indices: since $f_{sw}/f_g = 43.2$, they are interharmonics, mainly at $f_{sw} \pm 2f_g$ and $f_{sw} \pm 4f_g$ (2.2 and 1.6 % of I_L each) and near $2f_{sw}$, about 4.6 % of I_L in total up to 5 kHz, nearly independent of the load. With a grid-synchronized carrier they would exceed the 0.3 % limit, so an *LCL* filter is required at this switching frequency. PI control gives a TDD of 0.09–0.11 % and the same switching components within 0.1 % of I_L : this ripple is set by the modulation and the filter, not by the control.

Per sample, the network needs about 200 multiply–accumulate operations and ten tanh evaluations, and the flat and PBC laws fewer than 100 operations, i.e., a few MFLOP/s at 10 kHz, within the capability of current digital signal processors.

VII. CONCLUSION

A complete model of a 2 MW direct-drive PMSG wind energy conversion system was developed with machine data from the literature. Its control combines flatness for the speed and DC-link energy loops, passivity-based current control for both converters, with a cascade stability proof, and a normalized adaptive neural compensation with guaranteed bounded parameters. Compared with PI vector control tuned on the same poles, it tracks the MPPT speed 3.8 times more accurately under model errors and limits the DC-link

deviation during a 20 % grid dip to 3.6 V (11.6 V with PI and feedforward), at the price of larger current errors. The network reduces the speed error by 38 %. Future work will address a braking chopper and pitch control, an *LCL* filter, saturations in the stability analysis, and experimental validation.

CONFLICT OF INTEREST

The authors declare there are no conflicts of interest.

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